

## Studio 2 • Name the Pieces

OPIM 5641 • Business Decision Modeling  
by hand • pencil only • ~15 min

Name: \_\_\_\_\_ NetID: \_\_\_\_\_

Work it by hand, take a photo of your finished page, and upload it to HuskyCT.

**Flair's Furniture** makes inexpensive **tables** and **chairs**. Each table takes **3 hours** of carpentry and **2 hours** of painting. Each chair takes **4 hours** of carpentry and **1 hour** of painting. This month there are **2,400 carpentry hours** and **1,000 painting hours** available. Marketing wants **no more than 450 chairs** (there is already a pile of them) and **at least 100 tables** (inventory is low). Each table earns **\$7** profit; each chair earns **\$5**.

### 1 • Decision variables what do we get to choose?

### 2 • Objective function what are we making as big as possible?

$$\text{Max } Z = \text{ } T + \text{ } C$$

The two numbers you just wrote are the **objective function coefficients**. In one sentence, what does the 7 actually mean? \_\_\_\_\_

### 3 • Constraints what rules can we not break?

$$\text{ } T + \text{ } C \leq \text{ } (\text{ })$$

$$\text{ } T + \text{ } C \leq \text{ } (\text{ })$$

$$\text{ } \leq \text{ } (\text{ })$$

$$\text{ } \geq \text{ } (\text{ })$$

$$T, C \geq \text{ } (\text{ })$$